# ECA1011 - MICROPROCESSORS AND MICROCONTROLLERS

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## EXP-1: Analysis and development verification of 8-bit and 16-bit arthemetic operation by using assesmbly programming language.

## Objective:

**1.** 8-bit add and sub using indel 8-bit microprocessor.

**2.** 16-bit add and sub using indel 8-bit microprocessor.

**3**.implementation and testing of 8-bit multiplication and division using 8-bit microprocessor.

**4.** implementation and testing of 16-bit multiplication and division using 8-bit microprocessor.

## Components used:

8085-simulator (or) 8085 intel micro processor.

## Procedure:

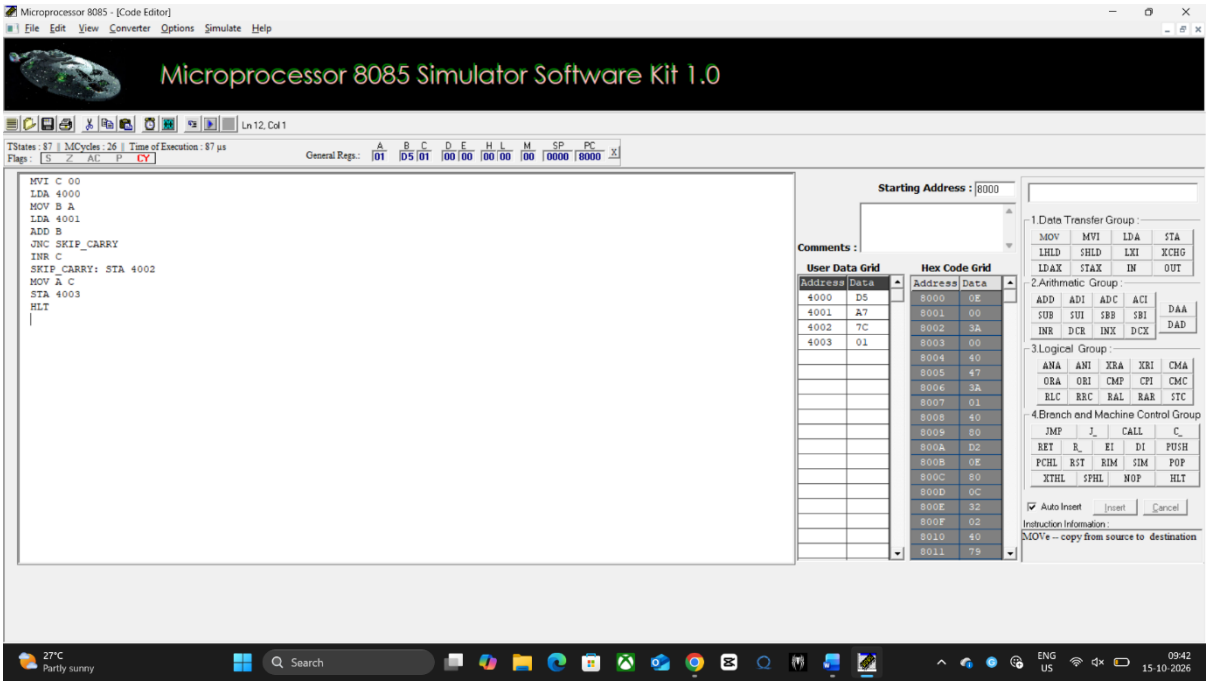
1. Open the 8085 Simulator on the computer.
2. Write the assembly language program for 8-bit addition and subtraction.
3. Write the assembly language program for 16-bit addition and subtraction.
4. Develop programs for 8-bit multiplication and division.
5. Develop programs for 16-bit multiplication and division.
6. Execute the programs using suitable input values.
7. Check the results and verify the output.

1. **To Analysis and Verify 8 bit Addition and Subtraction using 8 bit Micro Processor**

**Code:**

**Addition:**

```
LDA 4000
MOV B A
LDA 4001
MVI C 00
ADD B
JNC SKIP
  INR C
SKIP:
  STA 4002
  MOV A C
  STA 4003
  HLT
```



**Data Table:**

| | |
|---|---|
| **4000** | **D5** |
| **4001** | **A7** |
| **4002** | **7C** |
| **4003** | **01** |

**Subtraction:**

LDA 4000

MOV B A

LDA 4001

MVI C 00

SUB B

JNC SKIP

  INR C

  CMA

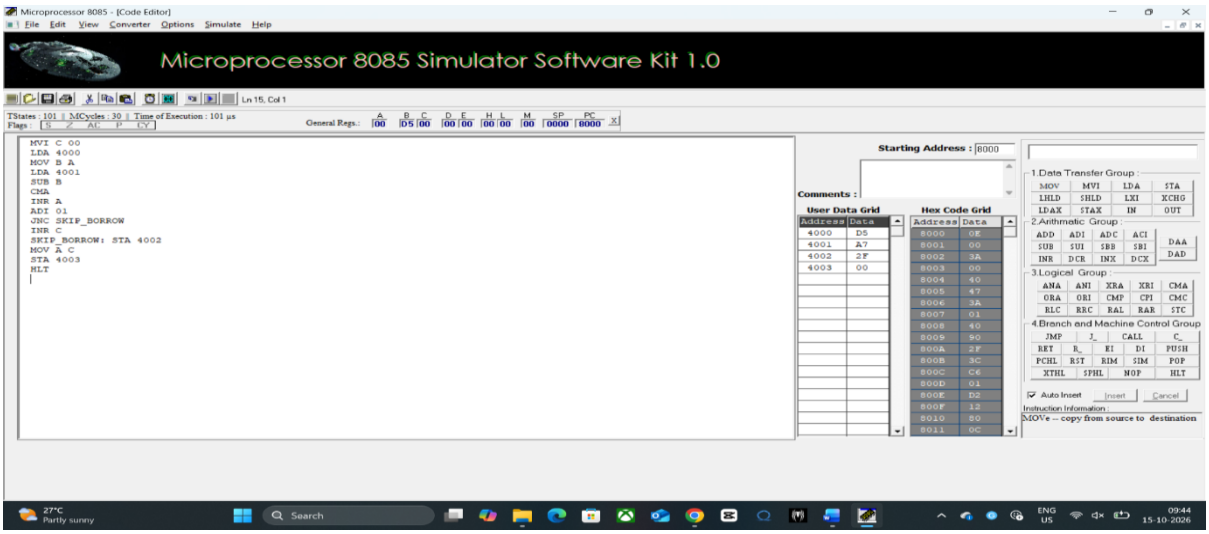
  ADI 01

SKIP: STA 4002

  MOV A C

  STA 4003

  HLT



**Data Table:**

| | |
|---|---|
| **4000** | **D5** |
| **4001** | **A7** |
| **4002** | **7C** |
| **4003** | **01** |

## 2. To Analysis and Verify 16 bit Addition and Subtraction using 8 bit Micro Processor

**Code:**

**Addition:**

MVI C 00

LHLD 4000

XCHG

LHLD 4002

DAD D

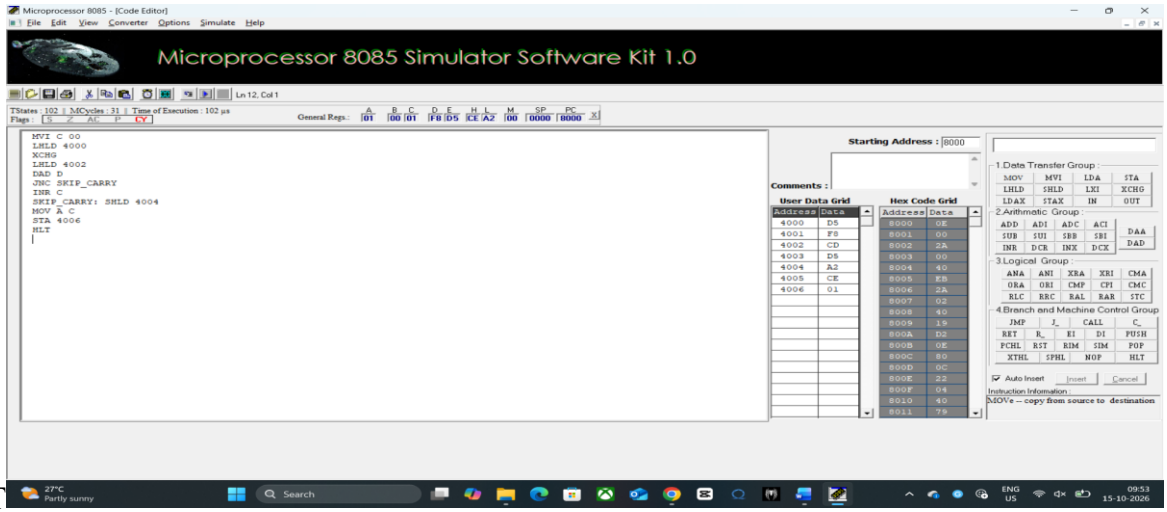
JNC SKIP_CARRY

INR C

SKIP_CARRY: SHLD 4004

MOV A C

STA 4006




HLT

**Data Table:**

| | |
|---|---|
| **4000** | **D5** |
| **4001** | **F8** |
| **4002** | **CD** |
| **4003** | **D5** |
| **4004** | **A2** |
| **4005** | **CD** |
| **4006** | **01** |

**Subtraction:**

MVI C 00

LHLD 4000

MOV A L

LHLD 4002

SUB L

MOV L A

MOV A H

SBB H

MOV H A

JNC SKIP_BORROW

INR C

CMA

MOV H A

MOV A L

CMA

ADI 01

MOV L A

MOV A H

ACI 00

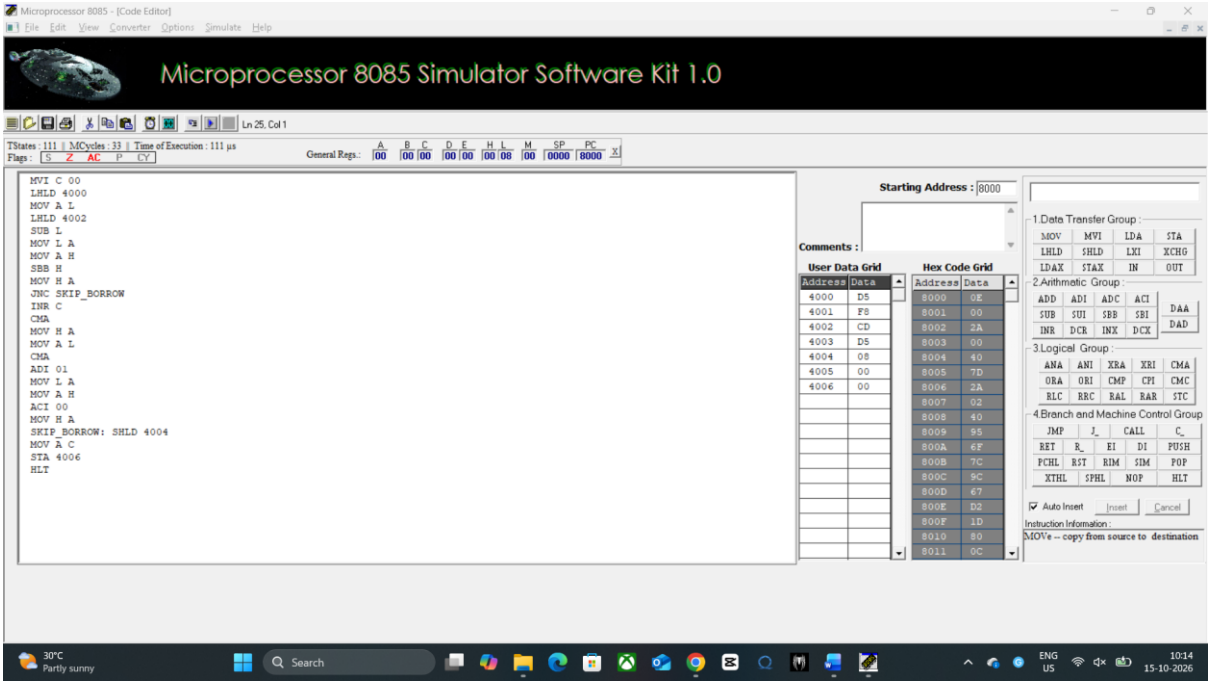
MOV H A

SKIP_BORROW: SHLD 4004

MOV A C

STA 4006

HLT




**Data Table:**

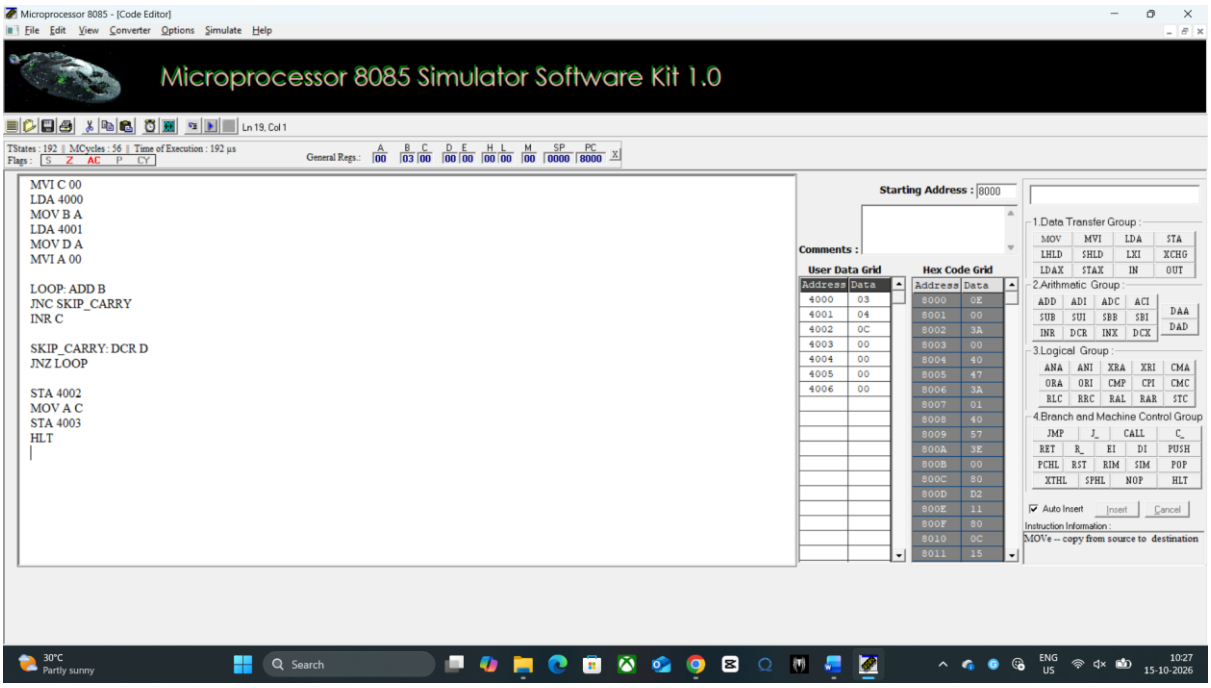
| | |
|---|---|
| **4000** | **D5** |
| **4001** | **F8** |
| **4002** | **CD** |
| **4003** | **D5** |
| **4004** | **08** |
| **4005** | **23** |
| **4006** | **00** |

3.**To Analysis and Verify 8 bit Multiplication and Division using 8 bit Micro Processor**

**Code:**

**Multiplication:**

```
MVI C 00
LDA 4000
MOV B A
LDA 4001
MOV D A
MVI A 00
LOOP: ADD B
JNC SKIP_CARRY
INR C
SKIP_CARRY: DCR D
JNZ LOOP
STA 4002
MOV A C
STA 4003
HLT
```



**Data Table:**

| | |
|---|---|
| **4000** | **03** |
| **4001** | **04** |
| **4002** | **0C** |
| **4003** | **00** |

**Division:**

MVI C 00

LDA 4000

LDA 4001

MOV B A

LDA 4000

LOOP: CMP B

JC STORE

SUB B

INR C

JMP LOOP

STORE: STA 4003

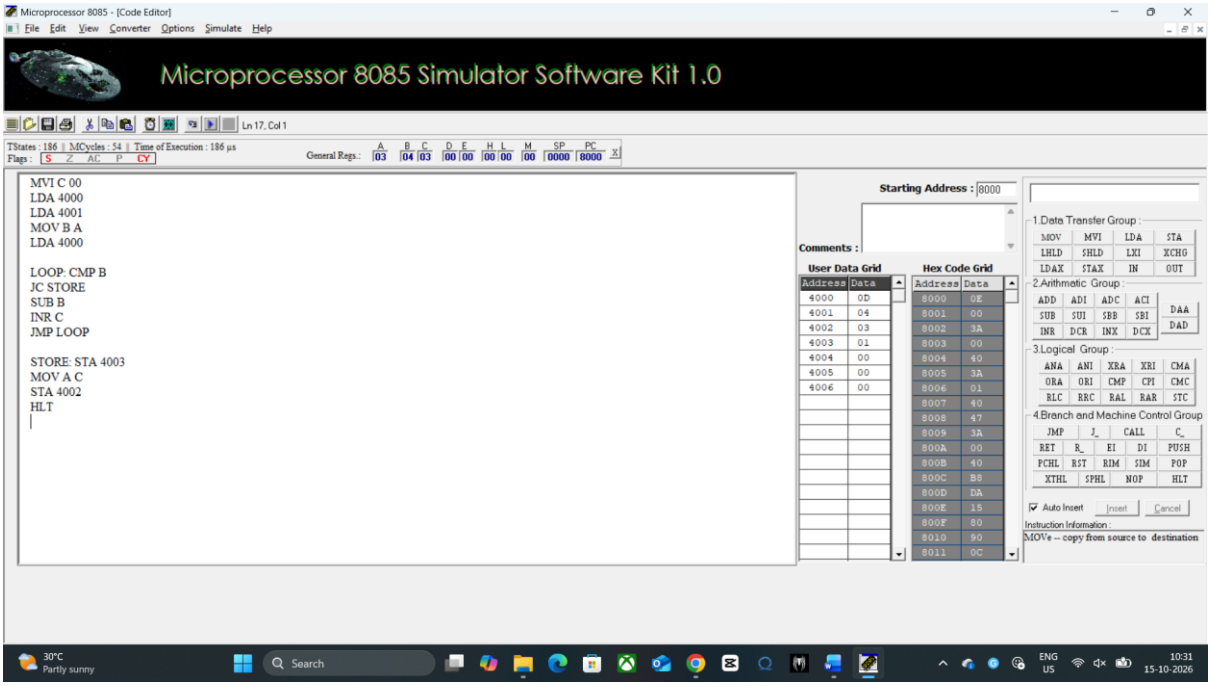
MOV A C

STA 4002

HLT

**Data Table:**

| | |
|---|---|
| **4000** | **0D** |
| **4001** | **04** |
| **4002** | **03** |
| **4003** | **01** |




**4.To Analysis and Verify 16 bit Multiplication and Division using 8 bit Micro Processor**

**Code:**

**Multiplication:**

LHLD 4000

XCHG

LHLD 4002

MOV C L

MOV B H

LXI H 0000

LOOP: MOV A C

ORA B

JZ STORE

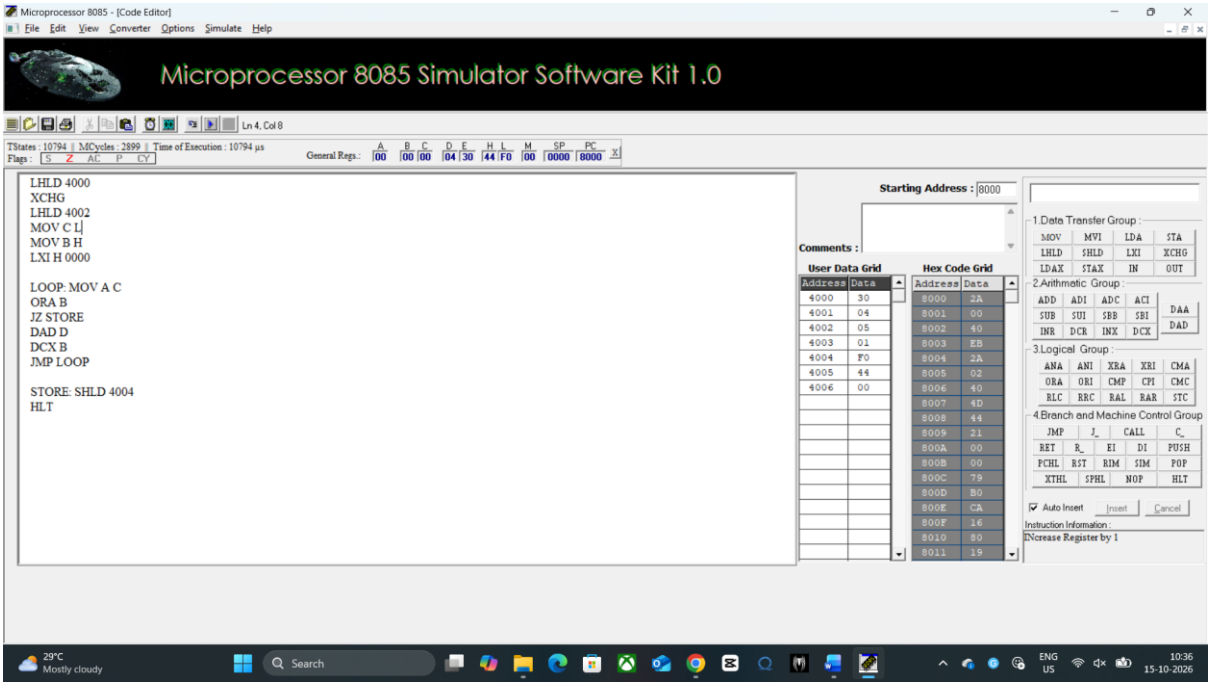
DAD D

DCX B

JMP LOOP

STORE: SHLD 4004

HLT




**Data Table:**

| | |
|---|---|
| **4000** | **03** |
| **4001** | **04** |
| **4002** | **05** |
| **4003** | **01** |
| **4004** | **F0** |
| **4005** | **44** |
| **4006** | **00** |

**Division:**

LXI B 0000

LHLD 4000

XCHG

LHLD 4002

LOOP: MOV A E

SUB L

MOV A D

SBB H

JC STORE

MOV A E

SUB L

MOV E A

MOV A D

SBB H

MOV D A

INX B

JMP LOOP

STORE: XCHG

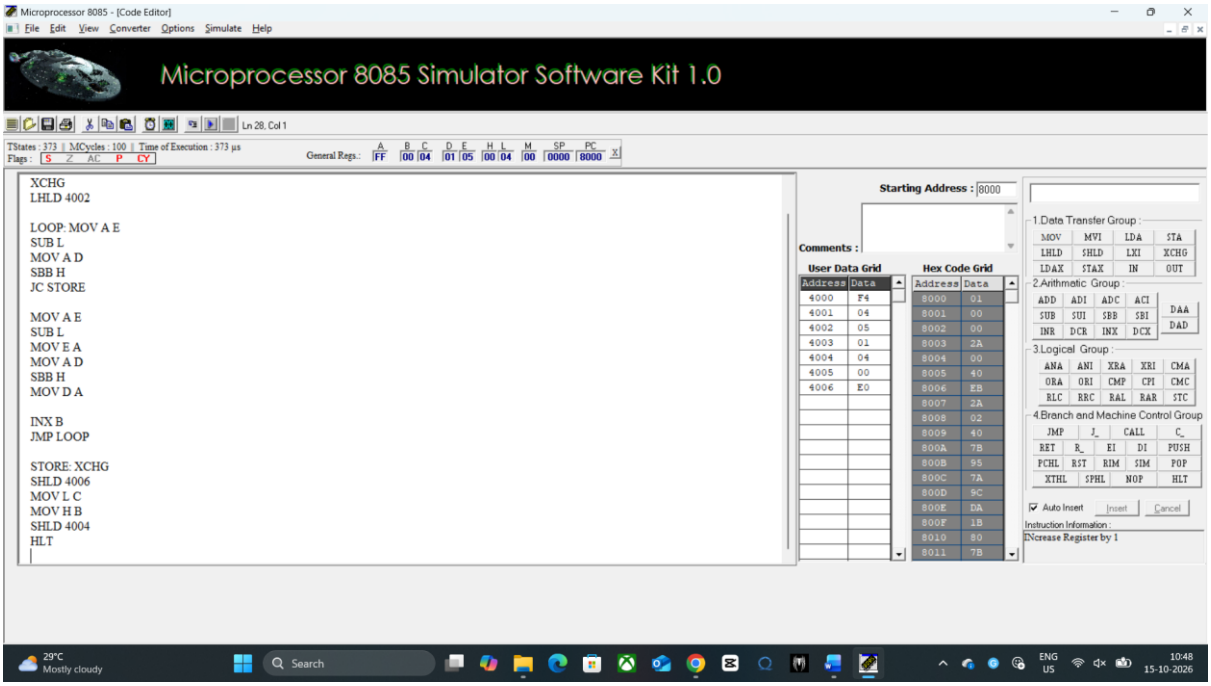
SHLD 4006

MOV L C

MOV H B

SHLD 4004

HLT



**Data Table:**

| | |
|---|---|
| **4000** | **F4** |
| **4001** | **3D** |
| **4002** | **05** |
| **4003** | **01** |
| **4004** | **3C** |
| **4005** | **00** |
| **4006** | **C8** |

**Result:**

1. 8-bit Addition and Subtraction: The target memory location holds the correct 8-bit sum or difference, and the adjacent byte records 01H if an overflow carry or borrow occurred, or 00H if it did not.
2. 16-bit Addition and Subtraction: The sequential memory addresses store the low and high bytes of the correct 16-bit sum or difference, along with a final byte reflecting the state of the terminal carry or borrow flag.
3. 8-bit Multiplication and Division: The output registers or memory addresses contain either a combined 16-bit product from repeated addition, or two distinct 8-bit bytes representing the exact Quotient and Remainder from repeated subtraction.
4. 16-bit Multiplication and Division: The designated target memory blocks accurately hold a multi-byte product (up to 32 bits) generated by the double-addition loop, or a pair of 16-bit values containing the final Quotient and Remainder.